

Land Survey

Engineering Mathematics II

Assignment 1 - Questions Only

Instructions: Answer all questions. Show formula, substitution, calculation, units and interpretation.

Section A: Formula and Concept Questions

1. Explain the meaning of the formula below and identify its main symbols. (4 marks)

$$\text{Gradient} = \text{Rise/Run}$$

2. Explain the meaning of the formula below and identify its main symbols. (4 marks)

$$\Delta E = L \sin \theta$$

3. Explain the meaning of the formula below and identify its main symbols. (4 marks)

$$\Delta N = L \cos \theta$$

4. Explain the meaning of the formula below and identify its main symbols. (4 marks)

$$\text{Area} = \frac{1}{2} |\sum x_i y_{i+1} - \sum y_i x_{i+1}|$$

5. Explain the meaning of the formula below and identify its main symbols. (4 marks)

$$d = \sqrt{(\Delta E)^2 + (\Delta N)^2}$$

Section B: Applied Questions

Problem Set 1

6. Solve a land survey problem involving levelling. Show formula, substitution, correct units and a technical interpretation. (10 marks)
7. Solve a land survey problem involving traverse computation. Show formula, substitution, correct units and a technical interpretation. (10 marks)
8. Solve a land survey problem involving plotting control lines. Show formula, substitution, correct units and a technical interpretation. (10 marks)
9. Solve a land survey problem involving area by coordinates. Show formula, substitution, correct units and a technical interpretation. (10 marks)
10. Solve a land survey problem involving survey closures. Show formula, substitution, correct units and a technical interpretation. (10 marks)

Problem Set 2

11. Solve a land survey problem involving levelling. Show formula, substitution, correct units and a technical interpretation. (10 marks)
12. Solve a land survey problem involving traverse computation. Show formula, substitution, correct units and a technical interpretation. (10 marks)
13. Solve a land survey problem involving plotting control lines. Show formula, substitution, correct units and a technical interpretation. (10 marks)
14. Solve a land survey problem involving area by coordinates. Show formula, substitution, correct units and a technical interpretation. (10 marks)
15. Solve a land survey problem involving survey closures. Show formula, substitution, correct units and a technical interpretation. (10 marks)

Problem Set 3

16. Solve a land survey problem involving levelling. Show formula, substitution, correct units and a technical interpretation. (10 marks)
17. Solve a land survey problem involving traverse computation. Show formula, substitution, correct units and a technical interpretation. (10 marks)
18. Solve a land survey problem involving plotting control lines. Show formula, substitution, correct units and a technical interpretation. (10 marks)
19. Solve a land survey problem involving area by coordinates. Show formula, substitution, correct units and a technical interpretation. (10 marks)
20. Solve a land survey problem involving survey closures. Show formula, substitution, correct units and a technical interpretation. (10 marks)

Problem Set 4

21. Solve a land survey problem involving levelling. Show formula, substitution, correct units and a technical interpretation. (10 marks)
22. Solve a land survey problem involving traverse computation. Show formula, substitution, correct units and a technical interpretation. (10 marks)
23. Solve a land survey problem involving plotting control lines. Show formula, substitution, correct units and a technical interpretation. (10 marks)
24. Solve a land survey problem involving area by coordinates. Show formula, substitution, correct units and a technical interpretation. (10 marks)
25. Solve a land survey problem involving survey closures. Show formula, substitution, correct units and a technical interpretation. (10 marks)

References and Further Reading

These notes are original learning materials prepared for Mr Mwirigi Mathematics Training Hub. They are academically informed by standard engineering mathematics texts and the local TVET curriculum framework. No textbook pages or worked examples have been copied.

1. Stroud, K. A. and Booth, D. J. Engineering Mathematics, 8th edition. Bloomsbury Academic.
2. Bird, J. Engineering Mathematics. Routledge / Taylor & Francis.
3. Bird, J. Higher Engineering Mathematics. Routledge / Taylor & Francis.
4. Relevant Civil Engineering, Building Technology and Land Survey curriculum and occupational standard documents.